

Two factor ANOVA under balanced design

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LSM

$$Y = [1_n : X_A : X_B : X_{AB}] \begin{bmatrix} \mu \\ \alpha \\ \beta \\ \gamma \end{bmatrix} + \varepsilon$$

$$1_n = 1_a \otimes 1_b \otimes 1_m$$

$$X_A = I_a \otimes 1_b \otimes 1_m$$

$$X_B = 1_a \otimes I_b \otimes 1_m$$

$$X_{AB} = I_a \otimes I_b \otimes 1_m$$

$$P_{1_n} = P_{1_a} \otimes P_{1_b} \otimes P_{1_m}$$

$$P_{X_A} = I_a \otimes P_{1_b} \otimes P_{1_m}$$

$$P_{X_B} = P_{1_a} \otimes I_b \otimes P_{1_m}$$

$$P_{X_{AB}} = I_a \otimes I_b \otimes P_{1_m}$$

$$P_{X_{AB}} = Q_1 + Q_A + Q_B + Q_{AB}$$

where $Q_1 = P_1$

$$Q_A = P_{X_A} - P_1$$

$$Q_B = P_{X_B} - P_1$$

$$Q_{AB} = P_{X_{AB}} - P_{X_A} - P_{X_B} + P_1$$

$$Q_1 Q_B$$

$$= Q_1 Q_A$$

$$= Q_A Q_B$$

$$= Q_{AB} Q_A$$

$$= Q_{AB} Q_B$$

$$= Q_{AB} Q_1$$

$$= 0$$

Constraints

$$1_n^T X_A \alpha = 0, 1_n^T X_B \beta = 0 \quad \text{--- } (*)$$

$$X_A^T X_{AB} \gamma = 0, X_B^T X_{AB} \gamma = 0$$

Transformed Design Matrix

Define

$$\tilde{X}_A = Q_A X_A$$

$$\tilde{X}_B = Q_B X_B$$

$$\tilde{X}_{AB} = Q_{AB} X_{AB}$$

$$(I - P_1) X_A = X_A - P_1 X_A$$

Then, $Q_A = P_{\tilde{X}_A}$

$$Q_B = P_{\tilde{X}_B}$$

$$Q_{AB} = P_{\tilde{X}_{AB}}$$

If (α, β, γ) satisfies (A) then

$$X_A \alpha = \tilde{X}_A \alpha$$

$$X_B \beta = \tilde{X}_B \beta$$

$$X_{AB} \gamma = \tilde{X}_{AB} \gamma$$

If α is s.t. $1^T X_A \alpha = 0$

$$\text{then } \tilde{X}_A \alpha = X_A \alpha - P_1 X_A \alpha$$

$$\underbrace{\frac{1}{n} 1 1^T X_A \alpha}_{=0}$$

$$Y = \mu 1 + X_A \alpha + X_B \beta + X_{AB} \gamma + \epsilon$$

$$= \mu 1 + \tilde{X}_A \alpha + \tilde{X}_B \beta + \tilde{X}_{AB} \gamma + \epsilon$$

mutually orthogonal

LS criterion

$$\begin{aligned} Q(\mu, \alpha, \beta, \gamma) &= \|Y - \mu 1 - X_A \alpha - X_B \beta - X_{AB} \gamma\|^2 \\ &= \|Y - \mu 1 - \tilde{X}_A \alpha - \tilde{X}_B \beta - \tilde{X}_{AB} \gamma\|^2 = ** \end{aligned}$$

Use the fact that $P_{X_{AB}} = Q_1 + Q_A + Q_B + Q_{AB}$

$$Y = (I - P_{X_{AB}}) Y + P_{X_{AB}} Y$$

$$= (I - P_{X_{AB}}) Y + (Q_1 Y + Q_A Y + Q_B Y + Q_{AB} Y)$$

orthogonal

$$\begin{aligned} ** &= \|(I - P_{X_{AB}}) Y\|^2 + \|(Q_1 Y - \mu 1) + (Q_A Y - \tilde{X}_A \alpha) \\ &\quad + (Q_B Y - \tilde{X}_B \beta) + (Q_{AB} Y - \tilde{X}_{AB} \gamma)\|^2 \\ &= \tilde{Q}(\mu, \alpha, \beta, \gamma) \end{aligned}$$

$$SSA = Y^T Q_A Y$$

$$SSB = Y^T Q_B Y$$

$$SSAB = Y^T Q_{AB} Y$$